HIKET — the storyline

a soil that carries its history: the trilemma of the balanced start, tested in Finland

Version of October 21, 2026. Numbers restated on 2026-10-13 on the recalibrated pair with the 1961–1984 pre-run climate: production `20261012_1042*`; balanced start `20261012_12*-13*`; S16 recalibrated on 2026-10-21 with the balanced start and the climate held, `20261013_13*`.

How to read this. A basis for the story discussion, not a draft. Each section opens with a strip of the whole paper, its own place highlighted, and then its internal arc. In the diagrams: blue = the section's payload; violet = a thread of the trilemma; amber fill = a caveat that travels with a claim. Figures sit where they are used: captions say *main text*, **appendix** or **not in your list: keep?**. Green boxes are the questions still open.

The decisions this storyline is built on (Lorenzo, 2026-10-12; story recomposed 2026-10-13 for the discussion with Samuli and Aleksi).

- **General first, then the case.** Soil carbon carries a long history, and the usual assumption of steady state does not capture it. Then the particular: Finland, where the forestry history clearly changed and the soil has been measured, and where the measurements show the accumulation slowing — HIKET's own question, *is the sink approaching saturation?*
- **The trilemma is introduced in Finland, as a hypothesis.** The kinetics, the inputs and the soil series cannot be reconciled under the steady-state assumption: motivated in the Introduction, tested in the Results. Published kinetics are a source of information like the others.
- **R1 opens the Results with the first test of the trilemma**: the steady state assumed in 1985 does not work. The tension, only hypothesised in the Introduction, becomes measured (F16).
- **R2 is the resolution**: the relaxed assumption works, and exposes the trilemma further — the ridge between inputs and kinetics shifts in model space and the optima move (F16 six panels, F12, F18).
- **The scenario decomposition closes R2 and answers HIKET's questions**: the assumptions carry a considerable weight, quantified (S16, every run its own calibration, so a comparison between assumptions only; S15 per model changes surprisingly little). It is the bridge to the forecast. Three parts for now; five terms once the two joint-effect runs are recalibrated with the new input data.
- **R3 asks the question the resolution raises**: are there consequences for forecasts — did we get it wrong so far? Two side arcs: **the sensitivity to temperature** under the relaxed assumption, and **the inputs**, central to HIKET — are we approaching saturation under the more realistic assumption, how much input would push it forward in time, with the short digression that saturation comes regardless and can only be postponed.
- **R4 is future development**, a side arm — relevant, but quite independent from the rest, an analysis of the implications: good trends, poor local skill; the residual analysis and its relationships; the hypotheses (understorey and pioneer vegetation missing), with literature mixed with our results; it ends with suggestions. The separate Discussion section is folded into it.
- **The load-bearing figures** are those sent to Samuli and Aleksi (email, 2026-10-13): Introduction F3_mean_SOCalt, F10b, F1 (S18 defending the inputs); R1–R2 F16 (six panels), F12, F18; the decomposition S16 (five terms; recalibrated three parts until the joint-effect runs exist); R3 F15 and the input table; R4 F11, F11 stand and soil, F11b. **Every other figure stays**, in the appendix; some can be promoted (marked in the figure plan).
- **The uncertainty stays a nested side arm** at the end of R1 (σ_{input} only) and of R2 (σ_{input} and σ_{init}): σ_{init}'s need was hidden in R1 by the assumption, not absent.
- **Conclusions, in this order**: (1) yes, we are approaching saturation; (2) the steady-state assumption shows its limits under a complex forestry history like Finland's; (3) this has consequences for forecasts; (4) the models need improving for the spatial variance.
- **The story follows the discovery.** We started from a general question set by HIKET and found things on the way, so the opening does not mirror the conclusions one to one. It still closes the loop: the saturation question opens the paper (the case) and is answered first (conclusion 1); the limit of the assumption is the obstacle found on the way.

The story on one page

0.0.0.1 The general problem. Soil carbon carries a long history. Its slowest pools adjust to a change in their input over centuries, much longer than the few decades a unit of carbon spends in the soil on average, so the soil of today still reflects the land use and the inputs of long before any model starts. The usual assumption, a start in balance with the present litter, does not capture that history.

0.0.0.2 The case. Finland: a forestry history that clearly changed — slash-and-burn, grazing and high-grading, the war decades, then a century of regrowth — and soil carbon measured three times on the same plots over four decades. The measurements show the soil gaining and the gain slowing. Is the forest soil sink approaching saturation?

0.0.0.3 The hypothesis: a trilemma. In Finland three sources of information are available on the same ground: the published *kinetics* of the inventory's own model family, the plot-level *inputs*, and the *soil series*. Under the steady-state assumption they cannot be reconciled: a model started in balance can satisfy two of them at most.

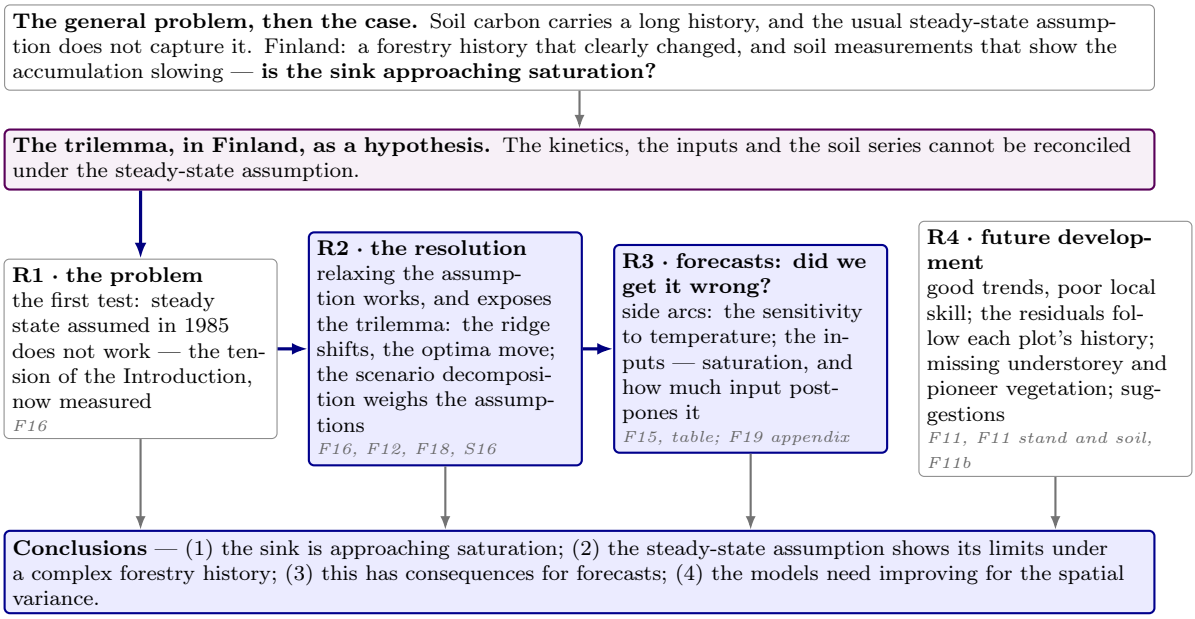
0.0.0.4 The test and the resolution. Started in balance in 1985 and recalibrated, all six models lose carbon over 1985–2024 against a measured gain. With the starting state estimated from the measurements and dated back to 1917, they reach the measured accumulation, and the ridge between input and turnover shifts: the optima move. The modelled sink is then the soil still catching up with its own history.

0.0.0.5 The weight of the assumptions. A scenario decomposition answers HIKET's questions: of the $+8.0\ \mathrm{tC\,ha^{-1}}$ the models gain over 1985–2024, the inherited state carries more than the whole, while the litter change since 1985 and the warming roughly cancel. The assumption about the starting state is not a detail of the method; it carries the sink.

0.0.0.6 The consequences. The starting state decides the forecast. A soil assumed balanced has no sink left; a soil carrying its history keeps one, which fades: yes, the sink is approaching saturation, while the stock stays well below balance. About a tenth more litter input would hold the sink of the past for twenty years, postponing saturation rather than removing it, and the response to warming shifts with the assumption.

0.0.0.7 Future development, a side arm. The models reproduce the national trend but not the differences between plots. What they leave behind follows the history of each stand, and points to inputs no tree-based product records: the understorey, natural mortality, the pioneer vegetation of young stands — and to some suggestions for the models.

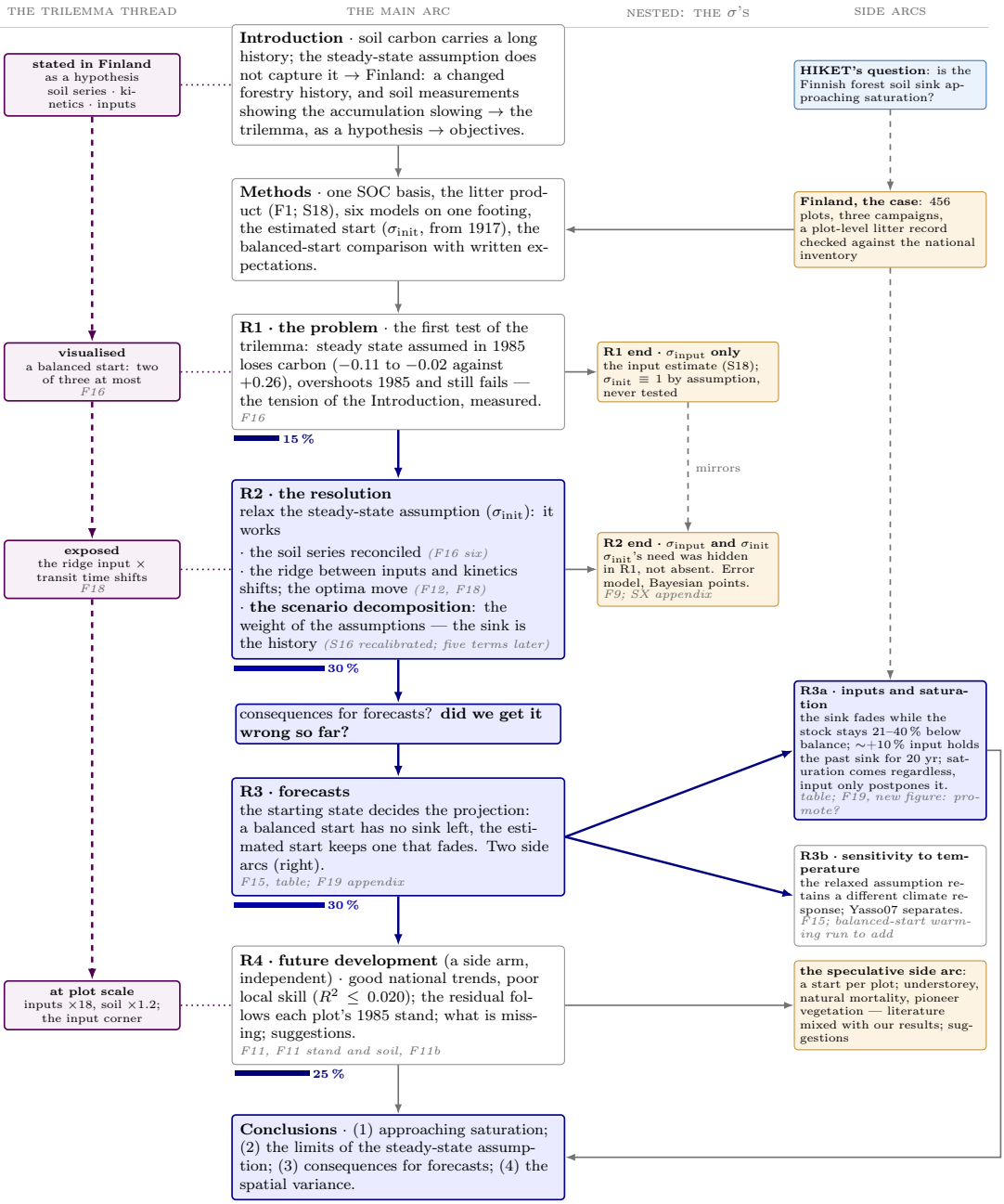
0.0.0.8 The arc of the whole paper.



The weight budget, proposed for the recomposed story: R1 the problem ∼15%; **R2** the resolution ∼30%; **R3** the forecasts and their two side arcs ∼30%; **R4** future development, with its speculative side arc ∼25%. R3 grows because the saturation question is conclusion 1.

The storyline schema

The whole paper on one page, in four lanes. **The main arc** runs down the middle: the Introduction moves from the general problem to the Finnish case and the hypothesis, R1 tests it, R2 resolves it by relaxing the steady-state assumption, and the resolution raises the question of R3 — did we get the forecast wrong? — before R4 turns to what the models still miss. **The trilemma thread** (left, violet) is stated in Finland as a hypothesis and comes back where it becomes visible: the failure of the balanced start (F16), the shifting ridge (F18), and the plots. **The nested σ side arm** closes R1 and R2. **The side arcs** (right) hang off R3 — inputs and saturation, the sensitivity to temperature — and off R4, the speculative arc on missing inputs. HIKET's saturation question opens the right lane and closes in conclusion 1. Bars give the proposed weight of each section.

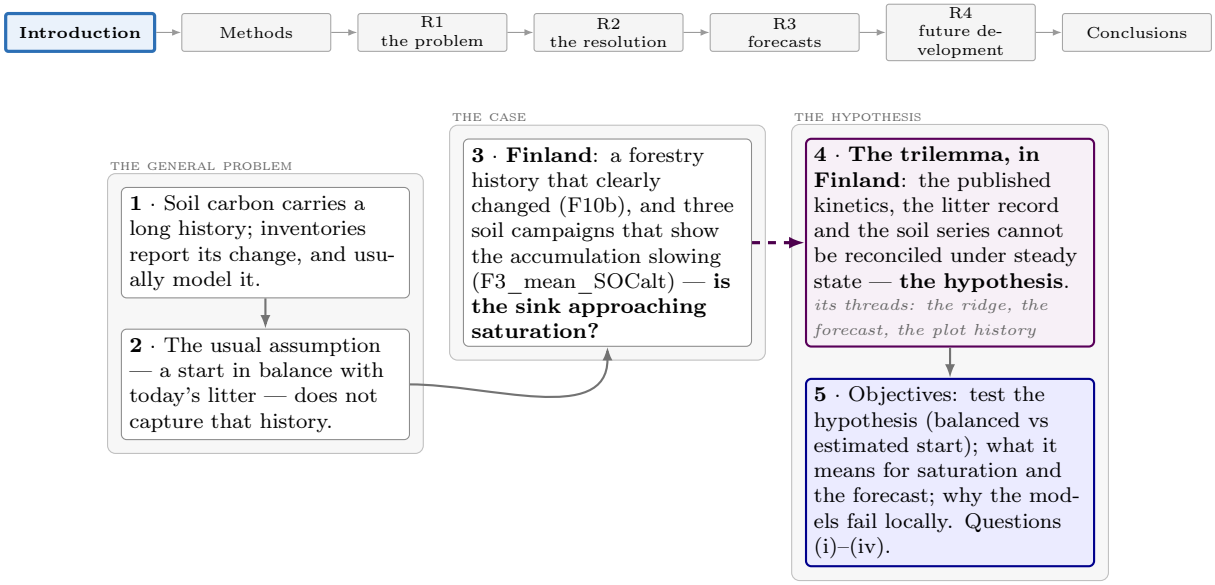


The figure plan

where	figure	role	status
Introduction	F3_mean_SOCalt	the case: the measured soil, gaining and slowing	**main**
Introduction	F10b_forest_history	the case: a forestry history that changed	**main**
Introduction	F1_inputs	the input corner of the trilemma	**main**
Introduction	*S18*	the inputs defended: plot product vs national inventory	appendix
Methods	*F2_baseline*	one homogenised SOC basis	appendix
Methods	*F4_initialization*	how the estimated start is built	appendix (promote?)
R1–R2	F16_eqinit_trajectories_six	the two assumptions against the campaigns, per model	**main**
R1	*F17*	the balanced start, rates by interval	appendix
R2	*F3_mean_soc*	the estimated start, six models together	appendix
R2	F12_mrt_yasso	transit times, calibrated and published	**main**
R2	F18_eqinit_tradeoff	the ridge shifts, the optima move	**main**
R2	*F14*	the ridge on the likelihood surface	appendix
R2	*F9*, *SX*	the two auxiliary parameters (nested part)	appendix (F9 promote?)
R2 end	S16_attribution_ensemble_recal	the scenario decomposition: the weight of the assumptions (*_joint* later)	**main**
R2 end	*S15_..._recal*	the same per model (changes little)	appendix
R3	F15_forward_arms	the response to warming	**main**
R3	*table*	the input that would hold the sink (political questions)	**main**
R3	*F19*	the forecast under the two assumptions: room left or not	appendix (promote?)
R3	*new*	saturation: held litter vs +20%, saturation postponed	to build (promote?)
R3	*new*	the warming response under the balanced-start calibration	to compute
R4	F11_rf_heatmap	what the residual follows	**main**
R4	F11_stand_and_soil	the stand of 1985 and the soil	**main**
R4	F11b_time_since_cut	the history of each stand: time since the cut	**main**
R4	*F6*, *harvest*, *Rh vs age*	skill; the speculative side arc	appendix

Main = the load-bearing figures sent to Samuli and Aleksi (14 with the table); **every other figure stays, in the appendix**, and some can be promoted: F19 (the forecast under the two assumptions, the clearest picture of "did we get it wrong?"), the saturation figure once built, F4, F9. Still open: (a) merge F10b and F3_mean_SOCalt? (b) R3b as framed needs the warming response of the *balanced-start* calibration — F15 compares published and calibrated parameters; a cheap rerun of the forward experiment.

1 Introduction



1.0.0.1 1. Soil carbon carries a long history, and its change is usually modelled. Inventories report the annual change in the soil carbon stock, not the stock. A few countries measure it (Sweden, Germany); most model it, Finland with Yasso07. The slow pools adjust over centuries, while the mean transit time is decades (Sierra et al. 2018): a soil keeps the memory of its past inputs for far longer than its turnover suggests.

1.0.0.2 2. The usual assumption does not capture it. Models are started in balance with a constant litter and climate — the Finnish inventory, the nationwide Yasso07 evaluation, RothC, the land models — and the inventory justifies it in one sentence ("approximately ten years of simulation since spin-up is enough to cancel out the effect of the spin-up level"). A balanced start responds to every change after its start date, but it cannot carry a deficit that already existed on that date: the site's history is replaced by an equilibrium with its present litter.

1.0.0.3 3. The case: Finland, and the question of saturation. A forestry history that clearly changed: slash-and-burn, grazing and high-grading, the war decades, then a century of regrowth (+84% growing stock) (Liski et al. 2006; Peltoniemi et al. 2004, 2007 said this in the group's own words). And the soil measured: three campaigns on the same plots, 64.5 → 71.5 → 73.6 $\mathrm{tC\,ha^{-1}}$, gaining and then slowing. The measurements suggest that the forest soil sink is approaching saturation — HIKET's question, and the one the paper answers first. ▶ F3_mean_SOCalt

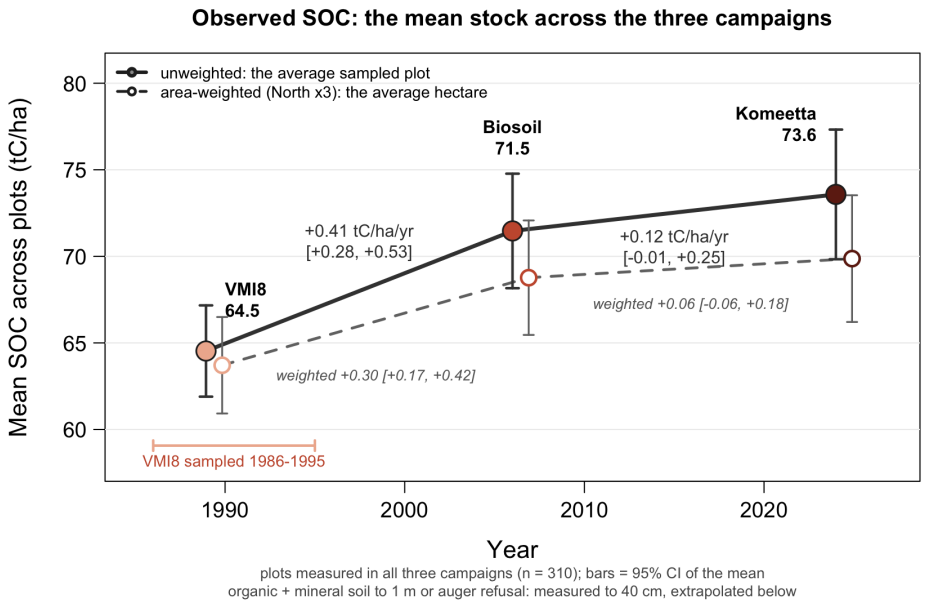


Figure 1: F3_mean_SOCalt — main text

▶ F10b_forest_history

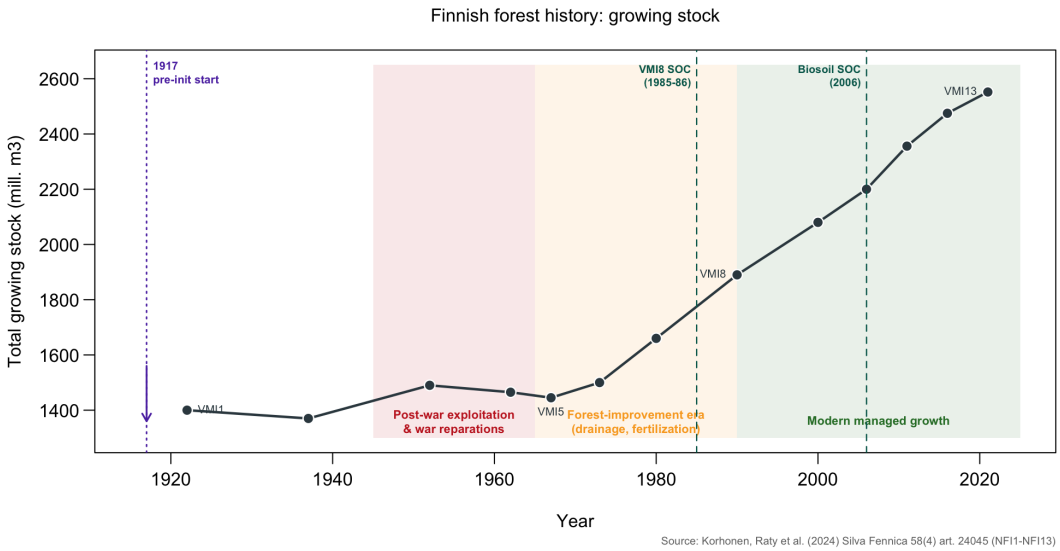
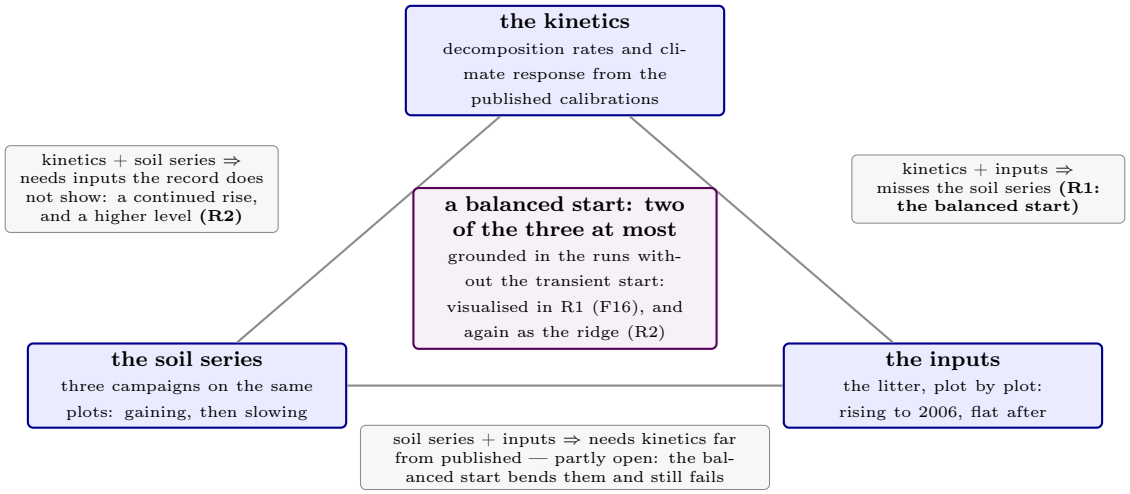


Figure 2: F10b_forest_history — main text

Two cautions. (a) The slowing rests on the 2006–2024 interval, the fragile one (its rate interval includes zero): "suggest", not "show". (b) The reported Finnish soil term turned into a source from 2021, once the litter input was recalculated; it can illustrate how a balanced start makes the soil term follow the litter. Describe the method, never judge the inventory (Aleksi leads it and is a coauthor): *the reported soil term is, in both directions, the response of a modelled soil to a litter input that departs from the level the soil was started in balance with.*

1.0.0.4 4. The trilemma, in Finland — the hypothesis. A model of a soil's trajectory has three sources of information to reconcile, and in Finland all three are measured on the same ground: the *soil series* (the three campaigns), the *kinetics* (the published calibrations of the inventory's own model family), and the *inputs* (a litter record for every plot, Tupek et al., rising by about 45% to 2006 and flat after, which agrees with the national inventory on the scope they share). In a soil in balance, the stock is the input times the transit time, so the three are tied together: a soil that keeps gaining carbon while its litter has stopped rising cannot be in balance with that litter. **Hypothesis: under the steady-state assumption the three cannot be reconciled** — a balanced start satisfies two at most. The trilemma rests on the balanced-start runs, which cannot appear in the Introduction; it is motivated here and tested at the opening of the Results (R1, F16). Its threads come back later: the ridge between input and turnover (R2), the forecast (R3), and the history of each plot (R4).



1.0.0.5 ⇝ The input corner is relatively robust. The litter estimates can be defended with a short discussion: on the scope it shares with the national inventory (tree turnover and harvest residues) the product agrees with it in the south, where most plots lie, and neither rises after about 2006; S18 in the appendix. ▶ F1_inputs

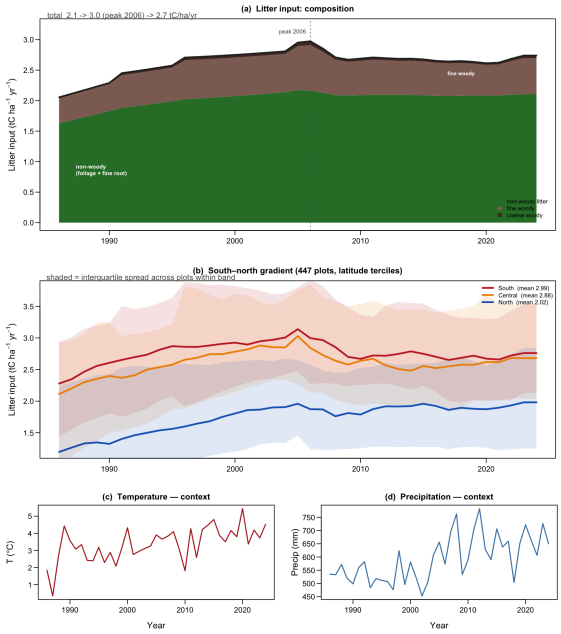
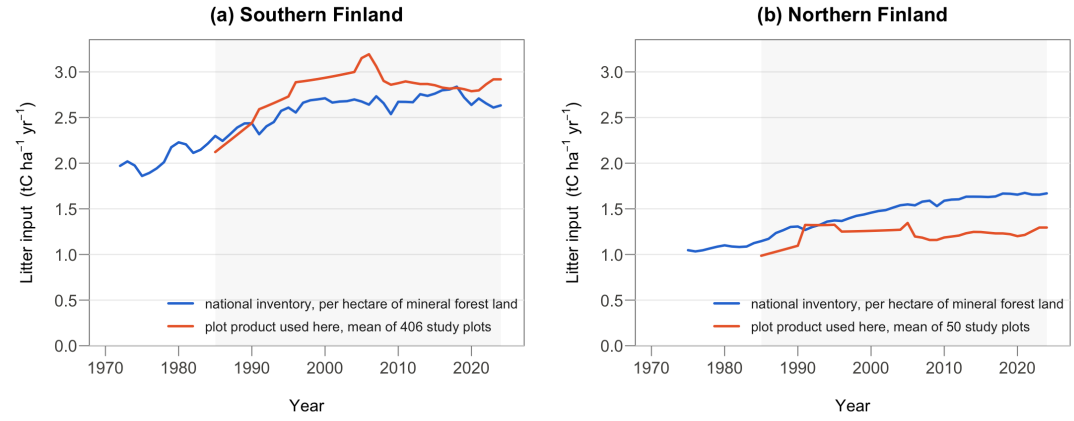


Figure 3: F1_inputs — main text

▶ S18_litter_vs_inventory



Both series: litter from the turnover of living trees and from harvest residues. The inventory's natural-mortality and ground-vegetation terms are excluded. Shaded: 1985-2024.

Figure 4: S18_litter_vs_inventory — **appendix**

1.0.0.6 5. Objectives. Test the hypothesis: the same six models, started in balance at the beginning of the measurements and with the starting state estimated and dated back to 1917, on the same data. Then follow the consequences to HIKET's question. Questions:

(i) Can a model started in balance reconcile the soil series, the kinetics and the inputs, and does relaxing the assumption?

(ii) Where does reconciling them place the models on the trade-off between litter input and transit time?

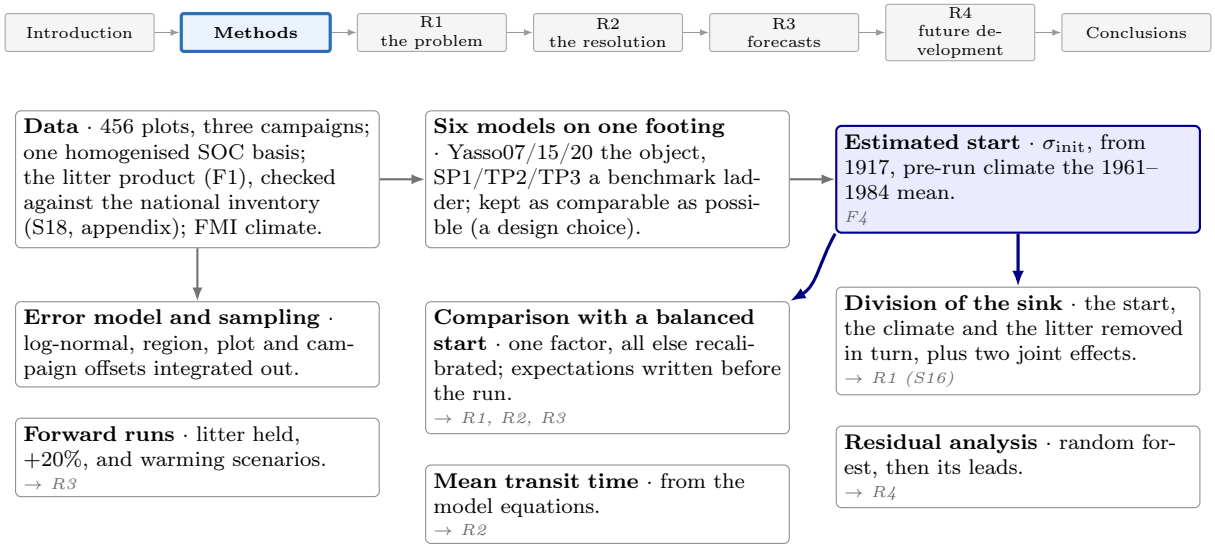
(iii) Under the relaxed assumption, is the forest soil sink approaching saturation, how does it respond to warming, and how much additional litter input would postpone saturation?

(iv) Why do the models reproduce the national trend but not the differences between plots?

Scope of the applied claim. The projections describe the calibrated models on the study plots, per hectare: the national inventory's term is not re-estimated, and the baseline projection holds the litter at 2024 and recycles the recent climate. The answer to (iii) is a statement about direction and sensitivity, not a forecast of the reported numbers. National units need a sourced area and Aleksi's review.

For the discussion. (a) The forecast consequence is now a question the Results raise (R3), not a claim of the Introduction; one sentence in paragraph 2 can still hint that a balanced start also decides what the model says about the future. (b) Name the balanced-start comparison in the objectives (as above), or only in the Methods?

2 Materials and methods



2.0.0.1 Data. As written. The litter input, one corner of the trilemma, is already shown in the Introduction (F1); here a short paragraph compares it with the national inventory's estimate (already in the draft): on the shared scope (tree turnover and harvest residues) the two agree in the south, where most plots lie, and neither rises after about 2006; the north differs, but holds only 50 of the 456 plots; the inventory also carries natural mortality, which grows and which the product lacks. S18 in the appendix. ▶ F2_baseline

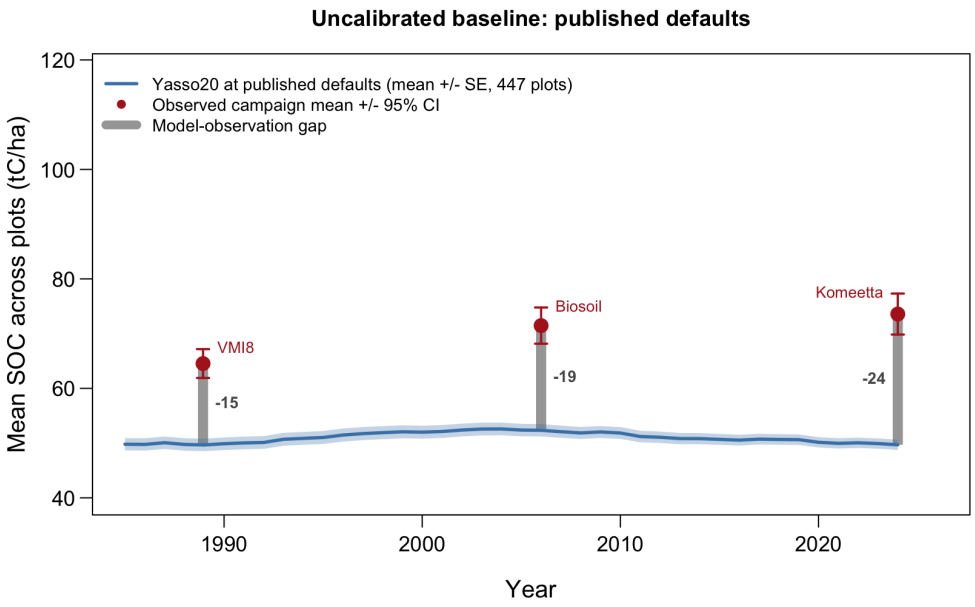


Figure 5: F2_baseline — **appendix**

2.0.0.2 The estimated start. As written, with the pre-run climate now the 1961–1984 mean, so that the soil enters 1985 adjusted to the climate it had lived in. ▶ F4_initialization

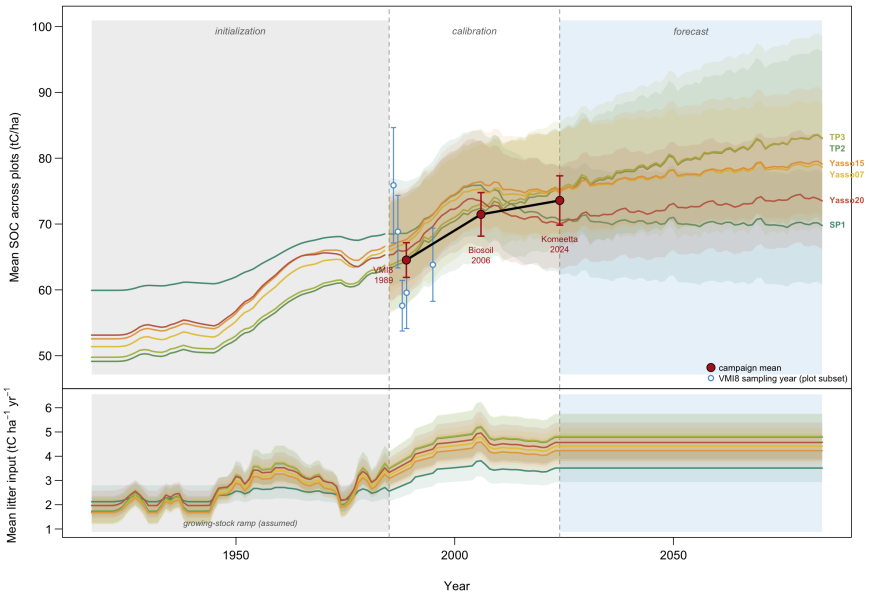


Figure 6: F4_initialization — **appendix**

2.0.0.3 The comparison with a balanced start. The same six models, data, likelihood and priors, with the start fixed at balance in 1985 (the 1985–1989 litter and the pre-run climate) and σ_{init} removed; everything else recalibrated. One factor. Stated as *the steady-state assumption applied at the start of the measurements*, the strongest fair version because the models are free to compensate, and explicitly not as the inventory's procedure. The expectations were written down before the run.

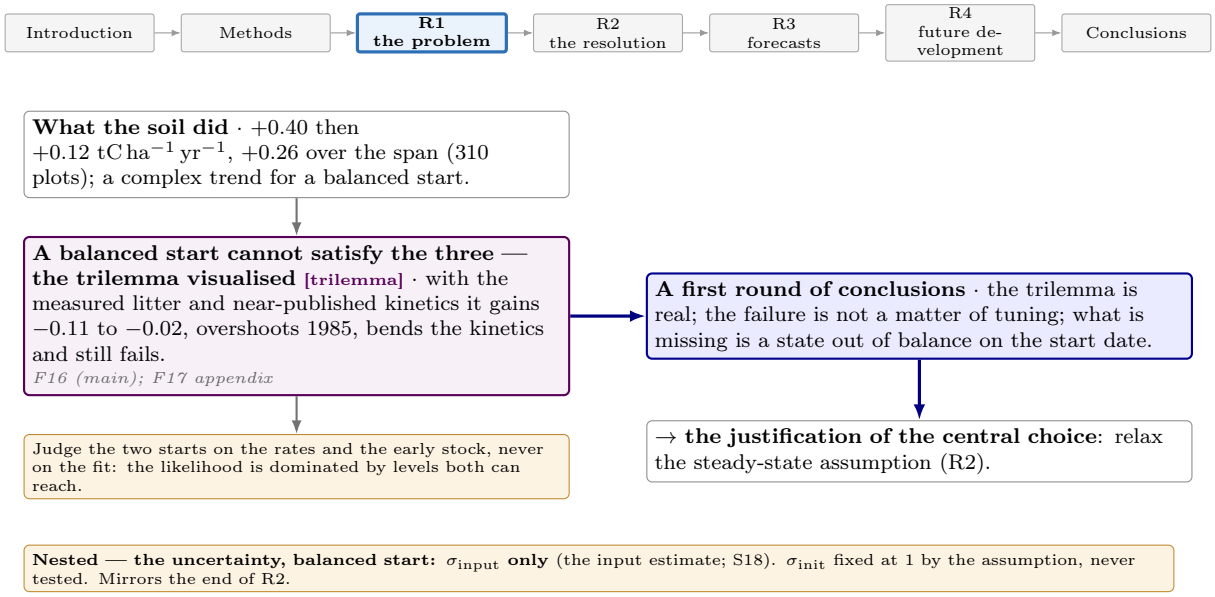
2.0.0.4 The division of the sink. Three runs and two joint effects of climate, in the words of the appendix.

> **Say in the Methods that the expectations were written before the run, and that two failed** (the balanced start overshoots the 1985 stock; its litter multiplier went down, not up)?

3 Results and discussion

R1 — THE OPENING: THE PROBLEM, TESTED THE HARD WAY

> **Role (Lorenzo, 2026-10-12).** R1 is the opening, a better definition of the problem. We know the conceptual problem of a balanced start, and we know the data hold a complex trend that could trouble it; here we test it the hard way and reach a first round of conclusions. From R1 to R2 runs the justification of the central choice.



3.0.1 What the soil did

3.0.1.1 +0.40 tC ha^{-1} yr^{-1} over the first interval, +0.12 over the second, +0.26 over the span (310 plots, unweighted, true sampling years). The second interval is the fragile one (its interval includes zero); the span is robust.

3.0.2 A balanced start cannot satisfy the three — the trilemma visualised

3.0.2.1 The first result, and the moment the trilemma of the Introduction becomes visible. Started in balance in 1985 and recalibrated, all six models lose carbon over 1985–2024, −0.11 to −0.02, against the observed +0.26. They rise with the litter to 2006 and give the rise back after it, and they start about 9 tC ha^{-1} above the first campaign (73–76 against 64.5), because a soil in balance with the 1985 litter holds more carbon than the soil held. Over 1985–2006 none reaches the observed rise (+0.06 to +0.20 against +0.40); over 2006–2024 all six lose carbon. F16 shows it against the campaigns: the two starts differ mainly at the first campaign, and separate again in the projection. ▶ F16_eqinit_trajectories_six

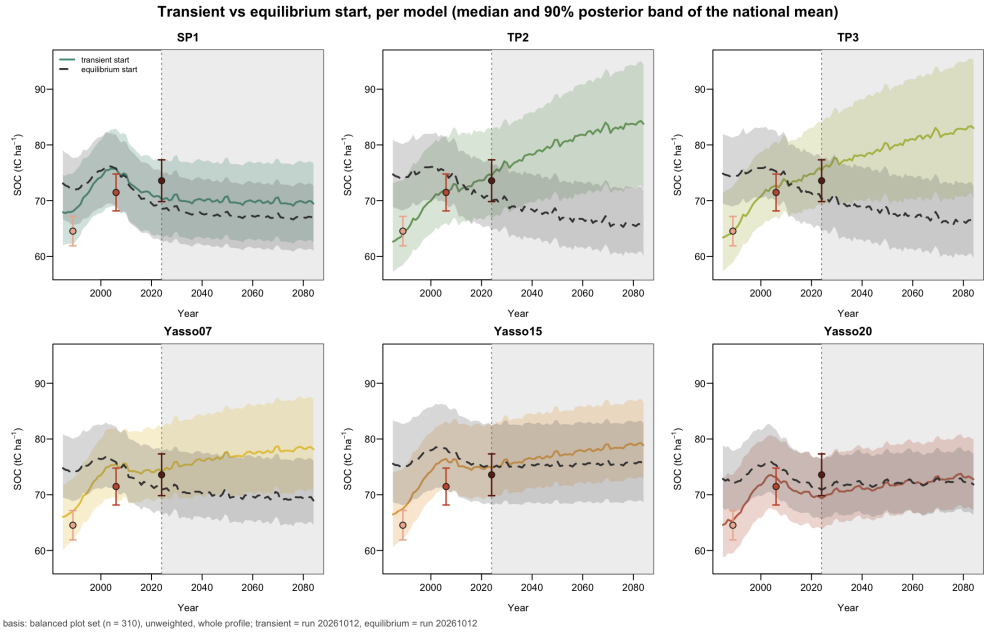


Figure 7: F16_eqinit_trajectories_six — main text

▶ F17_eqinit_rates

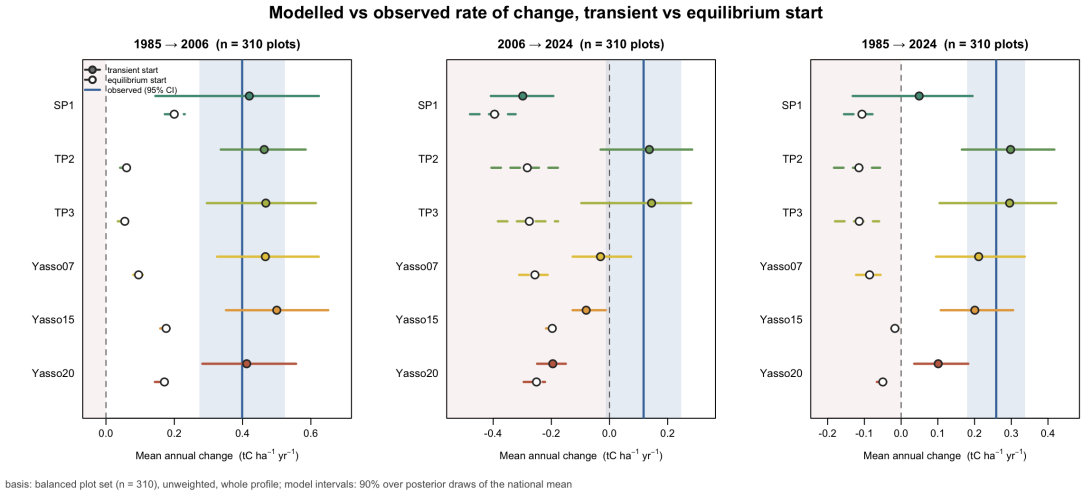


Figure 8: F17_eqinit_rates — **appendix**

3.0.2.2 ⇝ **The trilemma, contextualised.** The text names it here: held to the measured litter and to kinetics near the published ones, the balanced start gives up the soil series — the corner of the triangle the Introduction predicted. It is free to bend the kinetics and does — less input, faster and more temperature-sensitive decomposition (transit time 22–27 → 17–24 yr) — and still cannot produce the accumulation. How far the kinetics would have to move is the corner the comparison leaves partly open.

Do not judge the two starts by the fit. The likelihood is dominated by the stock levels, which both starts reach (log-likelihoods within 8, the estimated start better in five of six); the rates and the early stock discriminate. Say so in two sentences before a reader concludes the starts are equivalent.

Close the third corner? A balanced start with the rate priors widened would show how far the kinetics must move to fit both series; one factor, six jobs. Or state it as a limitation.

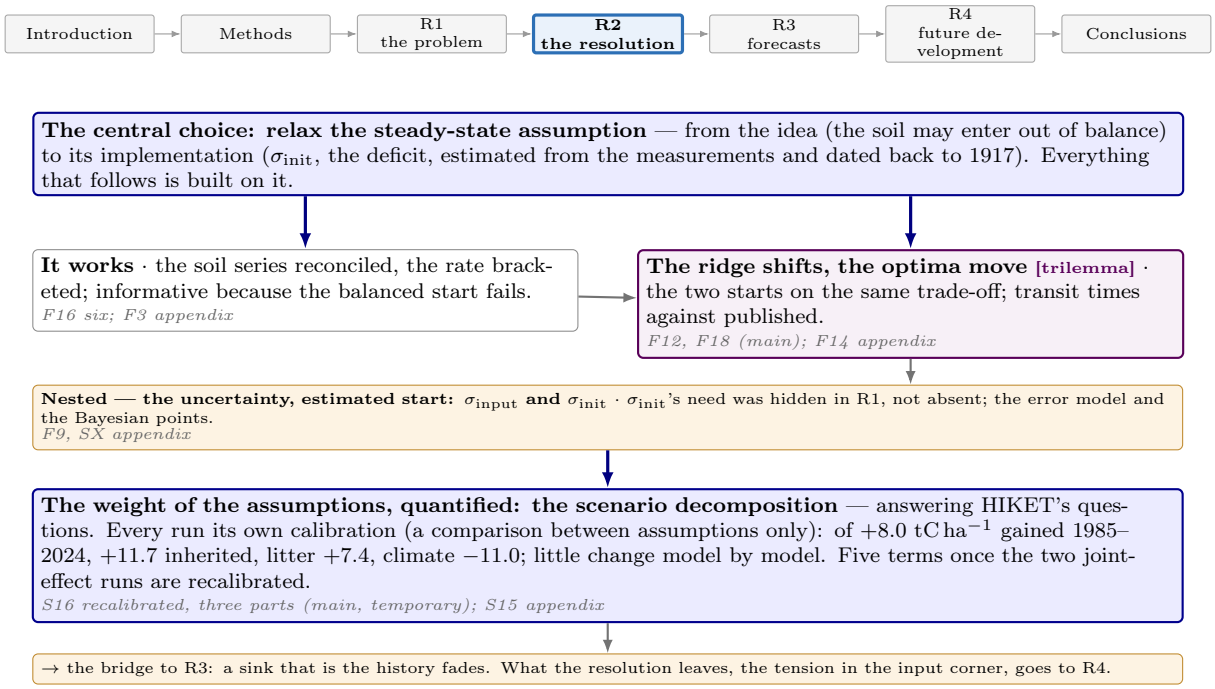
3.0.3 A first round of conclusions — and the justification of the central choice

3.0.3.1 Three things are settled before anything else is built. (1) The trilemma is real on these data: with the measured litter and near-published kinetics, a soil assumed balanced in 1985 cannot reproduce the measured accumulation. (2) The failure survives the freedom to compensate, so it is not a matter of tuning. (3) What the balanced start lacks is a state that is not in balance on the start date. **That is the justification for the central choice of the paper**: to relax the steady-state assumption and estimate the starting state from the measurements, which is what R2 does.

3.0.4 Nested: the uncertainty, as the balanced start sees it — σ_{input} only

3.0.4.1 The same side arm as at the end of R2, with one parameter. The balanced start carries one auxiliary uncertainty parameter, σ_{input}, which absorbs the uncertainty of the input estimate — the point where the data discussion (the litter product against the national inventory, S18) enters the calibration. In the balanced start it relaxes (Yasso15 1.54 → 1.31, TP2 1.75 → 1.60), while the kinetics strain. σ_{init} **is not absent: it is fixed at** 1 **by the steady-state assumption** — a value never questioned and never tested, only assumed. Implementation details here: the same error model and priors as R2, the fit equivalence of the two starts (the likelihood dominated by levels) and why the comparison is made on the rates.

R2 — THE RESOLUTION: RELAXING THE STEADY-STATE ASSUMPTION



Role (Lorenzo, 2026-10-12). The main arc, carrying the most load. Relaxing the steady-state assumption is the paper's own approach to the trilemma — the proposed solution, and *the central choice*. It moves the models across the ridge in parameter space, and the trade-off translates according to the new assumption. Everything that follows (the forecast, the plot analyses) is built on it, once it is justified; its ramifications are examined, but the choice is the centre.

3.0.5 The central choice

3.0.5.1 From the general idea to its implementation. The idea: the soil may enter the observations out of balance. The implementation: the start is estimated from the measurements and dated back to 1917, through one quantity, σ_{init}, the deficit the soil carries — the parameter the balanced start had fixed at 1. The same data, priors and likelihood. What σ_{init} means, and what it costs, is in the nested part at the end of R2.

3.0.6 The estimated start reconciles the soil series

3.0.6.1 All six reach the three campaign levels. Over 1985–2024 they span +0.05 to +0.30 against the observed +0.26, four of six inside its interval (SP1 +0.05, Yasso20 +0.10 below). Over 1985–2006 all six reach the observed rise; over 2006–2024 only TP2 and TP3 reach the observed +0.12, Yasso07 is near zero (−0.03), and SP1, Yasso15 and Yasso20 lose carbon. F16 (six panels) already shows the estimated start against the campaigns, model by model; F3, the six models together, goes to the appendix. ▶ F3_mean_soc

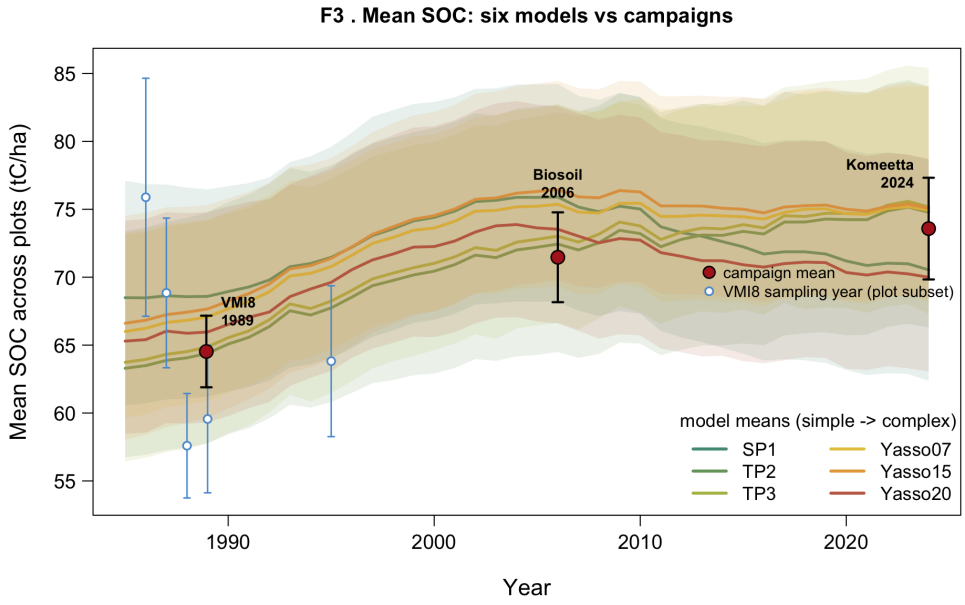


Figure 9: F3_mean_soc — **appendix**

3.0.6.2 ⇝ **Bracketing is informative because it is not automatic.** Matching the rate would mean little if any calibration on the same data could do it. The balanced start, calibrated on the same measurements with one degree of freedom fewer, fails in every model.

3.0.7 Where the models land: shorter transit times than published

3.0.7.1 Calibrated against published: Yasso07 and Yasso15 turn over faster, Yasso20 sits on its published value. ▶ F12_mrt_yasso



Figure 10: F12_mrt_yasso — main text

3.0.8 Across the ridge — the trilemma resolved as far as the data allow

3.0.8.1 The soil fixes the product of input and transit time, and the posteriors ride that trade-off. The two starts land at different places on it: in five of the six models the balanced start sits at less input and faster turnover (SP1, a single slow pool, barely moves). ▶ F18_eqinit_tradeoff

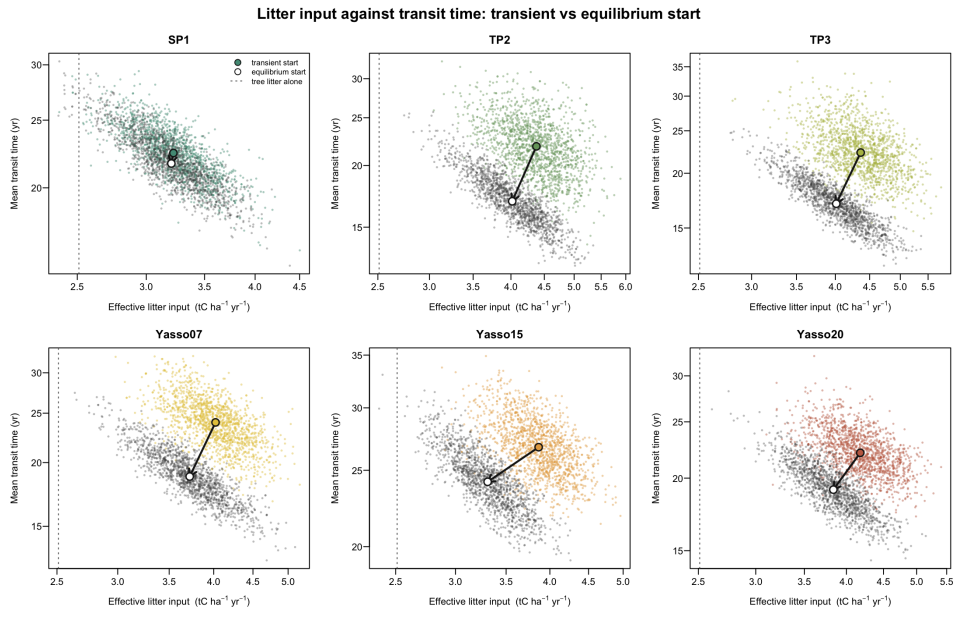


Figure 11: F18_eqinit_tradeoff — main text

▶ F14_mrt_ridge

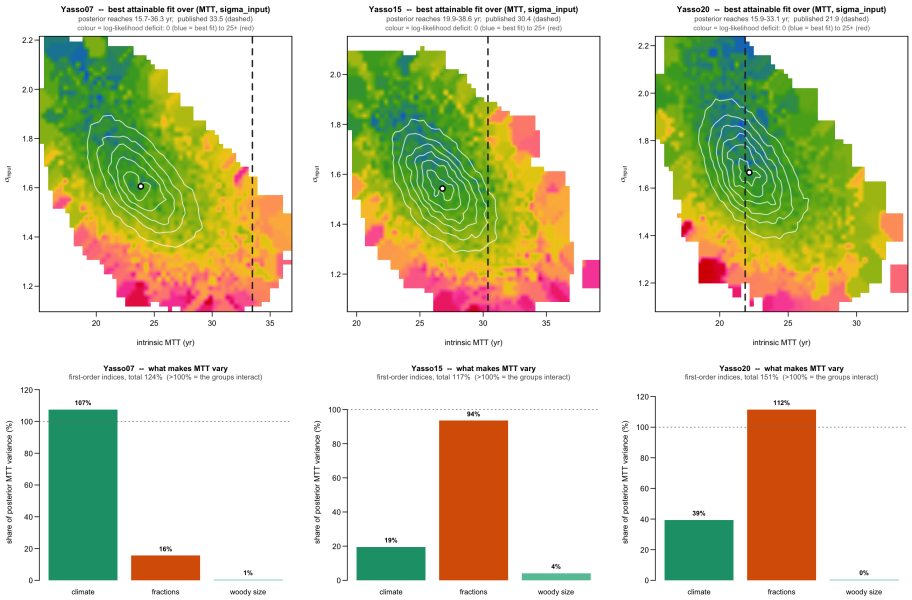


Figure 12: F14_mrt_ridge — **appendix**

3.0.8.2 ⇝ **The echo, in parameter space. The triangle comes back once more**, after F16 showed it in the trajectories. The soil series is the ridge; the measured input (the dotted line) and the published kinetics sit off it; a balanced start can stand on at most two of the three, and the estimated start, carrying the history, stands somewhere else on the same ridge. Relaxing the assumption moves the models across it, and the trade-off translates according to the new assumption. The text should say it in one or two sentences — after all the work, the trilemma is visible — without redrawing the

figure.

3.0.8.3 ⇝ **What the resolution leaves.** The tension is not removed but concentrated in one corner: the soil series is met and the kinetics stay near the published values (Yasso20 on its own, Yasso07 and Yasso15 faster), while the input the models need stays above what the product supplies. That remaining gap is the thread to R4 and its side arc on the inputs the product may miss.

3.0.9 Nested: the uncertainty, as the estimated start sees it — σ_{input} and σ_{init}

3.0.9.1 The same side arm as at the end of R1, now with two parameters. σ_{init} becomes needed, and **that need was there before**: in R1 it was hidden by forcing the steady-state assumption, which fixes it at 1. R2 may look more uncertain, but it only makes explicit a quantity R1 assumed without testing. The two auxiliary parameters share a width by design.

- σ_{input} — more input than measured: effective flux 3.22–4.40 against 2.511, inside the envelope of boreal production, set by the likelihood rather than the prior (the posterior stays above the prior centre whichever anchor is used). The national inventory's full estimate is 1.08–1.26× the product on the southern plots, bracketing the understorey-corrected anchor (1.21) and below the posteriors (1.28–1.75); SX sets it against independent estimates.
- σ_{init} — the deficit the data demand: 0.45–0.56 in five of six models (SP1 0.78), lower than the growing-stock record implies (~ 0.90), so read as an *effective starting deficit of the soil* (slash-and-burn, raking, heavy cutting) rather than a ratio of litter inputs. It carries the forecast: headroom correlates with it at -0.76 to -0.97 (R3).
- **The implementation, and its advantages**: the error model (region, plot and campaign offsets integrated out, a declared total of 0.800), what it buys (an honest number of independent observations, 221), and the Bayesian points — priors as information, homogenised across models; the posterior against the prior.

▶ F9_auxiliary_sigmas

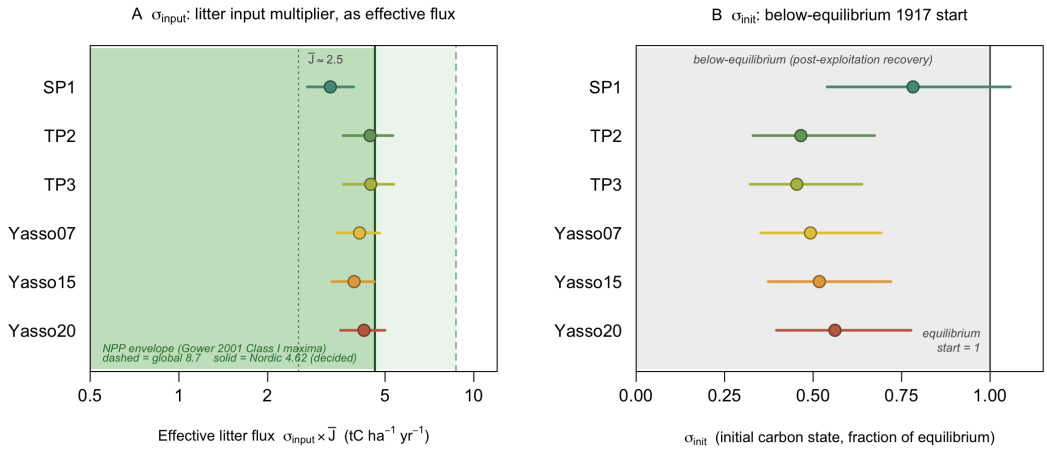


Figure 13: F9_auxiliary_sigmas — **appendix**

▶ SX_input_vs_literature

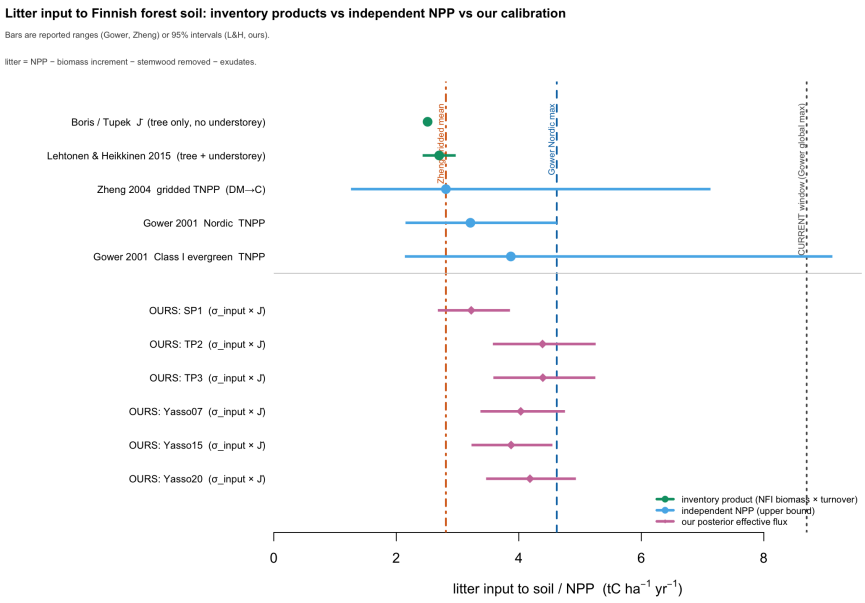


Figure 14: SX_input_vs_literature — **appendix**

The direction depends on the comparator, so the title names none. Against the published parameters the calibration has faster turnover (Yasso07, Yasso15); against the balanced start, slower. "Displaces along" covers both; never add a direction.

3.0.10 The weight of the assumptions, quantified: the scenario decomposition

3.0.10.1 Answering HIKET's questions. The scenario decomposition divides the modelled sink of 1985–2024 among what the soil inherited from before 1985, the litter change since 1985 and the climate. Every scenario carries its own calibration (free start; balanced start; balanced start with the climate held at the reference), so the division is a comparison between assumptions only. The assumptions carry a considerable weight, and this is it, quantified. Six-model average (medians): of +8.0 tC ha^{-1} gained, the inherited state carries +11.7; the litter change adds +7.4; the climate removes −11.0. The inherited part exceeds the whole sink in all six models, because the balanced-start calibration loses carbon in every draw: the litter change and the climate together remove a little carbon (−1.0 to −5.1). Shown model by model (S15), the picture changes surprisingly little. ▶ S16_attribution_ensemble_recal

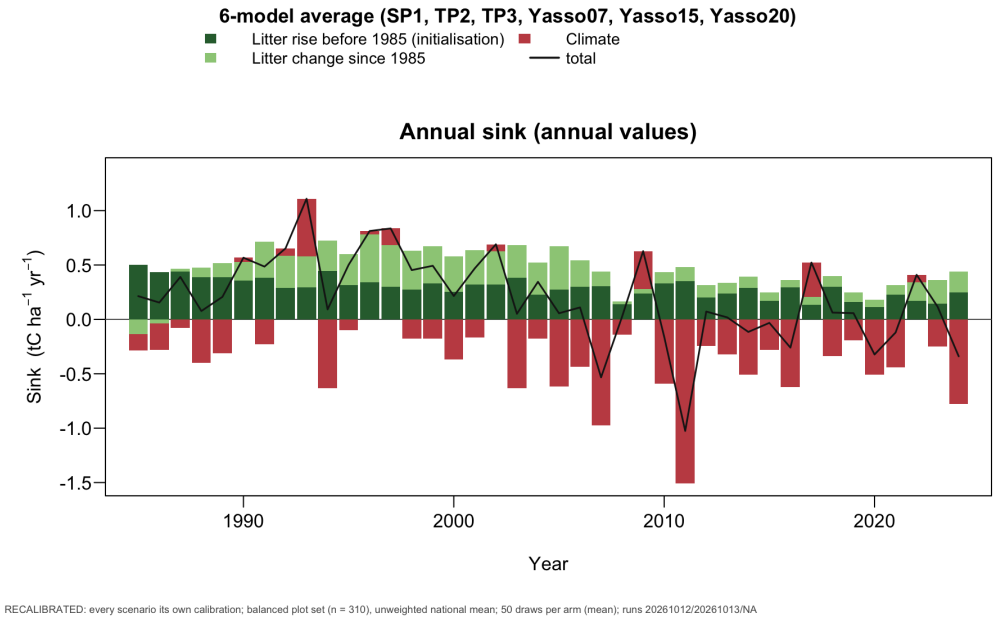


Figure 15: S16_attribution_ensemble_recal — main text

▶ S15_attribution_by_model_recal

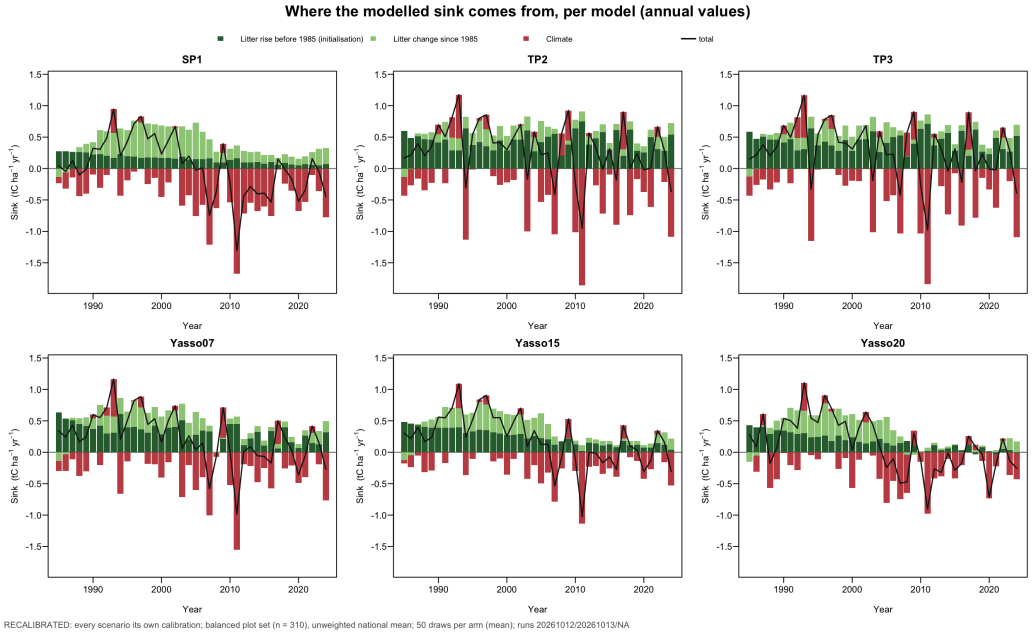


Figure 16: S15_attribution_by_model_recal — **appendix**

Three parts for now, the joint effects inside them by the order of the runs; the five-term version (joint effects +1.5 and −1.3 in the fixed-parameter division) waits until the two runs that measure them are recalibrated, with the new input data.

3.0.10.2 ⇝ **Necessary, and dominant.** The balanced start shows that the history is *necessary* — with everything else free, the models cannot do without it. The decomposition shows how much it

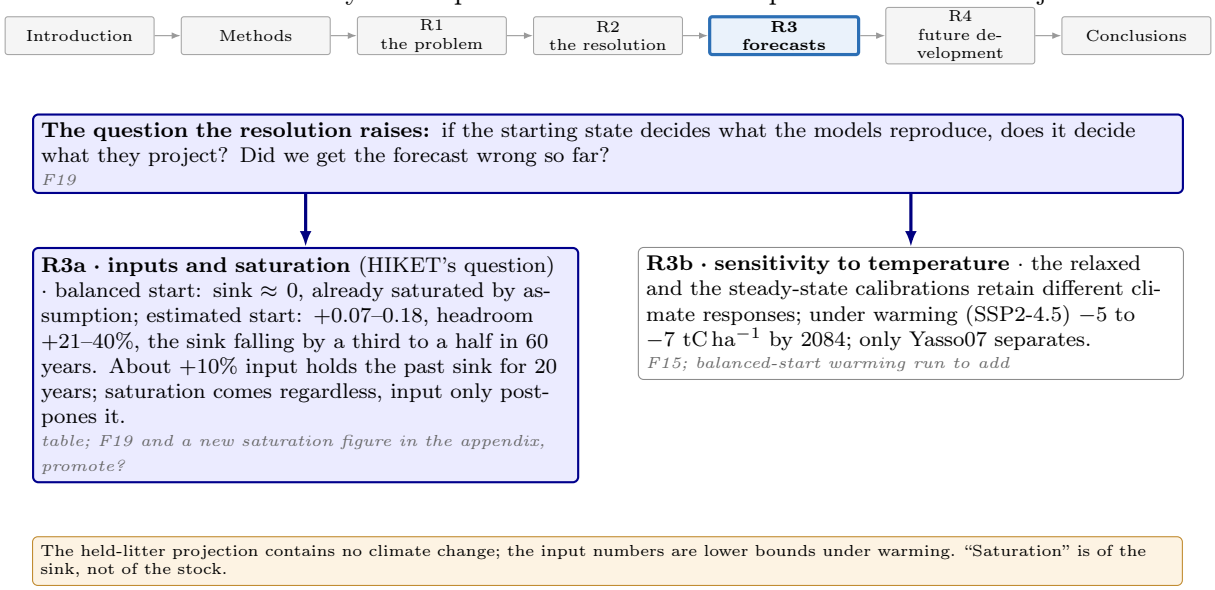
weighs — more than the whole sink, because the warming takes back more than the litter rise adds. The second answers the objection to the first ("the inherited part is large only because the calibration chose a low σ_{init}"): the data demand the deficit. **And it is the bridge to R3**: a sink that is the history is a sink that fades — the mechanism of the saturation the forecast will show.

3.0.10.3 ⇝ **Recalibrated: what changed.** Against the fixed-parameter division (one parameter set per model across the runs; appendix), the inherited part moves by 3% in the six-model mean (+11.3 → +11.7), the litter part shrinks by 9–24% (+8.9 → +7.4) and the climate part by about 10% (−12.2 → −11.0): runs calibrated with a balanced start retain less litter input, and less carbon passes through their soil. No part changes sign or rank, as expected before the run. The recalibrated version is more robust and easier to explain alongside the rest of the methods; the fixed-parameter one stays in the appendix (decide).

> **The size of the climate part depends on the reference climate** (1961–1984): it grew from −7.5 to −12.2 in the fixed-parameter three-part reading when the reference moved there. The robust statement is the ordering — the sink inherited; litter and warming of the same order and opposite sign. Say so in the text.

R3 — CONSEQUENCES FOR FORECASTS: DID WE GET IT WRONG?

3.0.10.4 The question the resolution raises. If the starting state decides what the models reproduce, it may decide what they project: did we get the forecast wrong so far? Tested on the same models, now projected forward, with two side arcs: the inputs and saturation — HIKET's question — and the sensitivity to temperature. Within the scope declared in the objectives.



3.0.11 The starting state decides the forecast

3.0.11.1 Projected over 2025–2044 with the litter held at 2024 and the recent climate recycled, the balanced start gives a sink of −0.12 to +0.02 tC ha^{-1} yr^{-1} and a headroom of −8 to +9%: a soil assumed balanced is saturated by assumption. The estimated start gives +0.07 to +0.18 and +21 to +40% (SP1 ≈ 0) — the disequilibrium Liski et al. (2006) found by another route (+38%). S16 explains it: the sink the models carry is the history, and the history fades. ▶ F19_eqinit_forecast

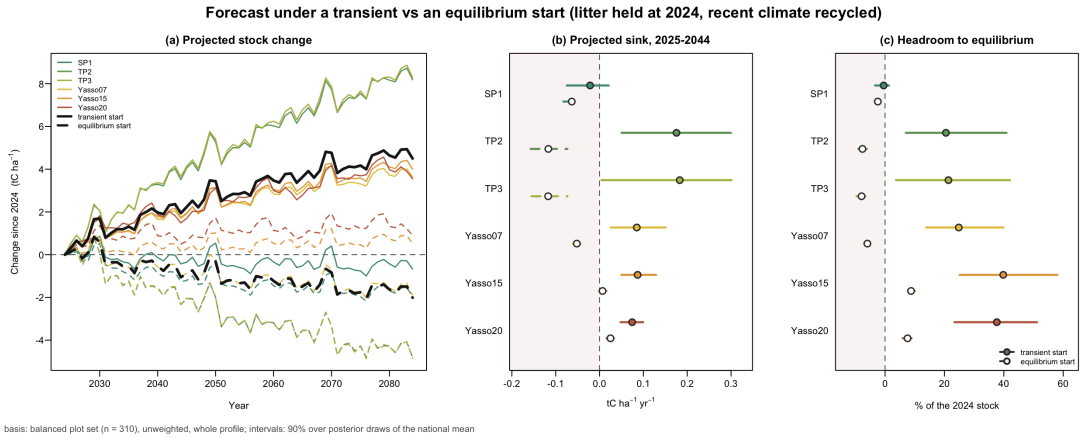


Figure 17: F19_eqinit_forecast — **appendix**

3.0.12 Side arc a: inputs and saturation — are we approaching it?

3.0.12.1 Yes, and the qualifier matters. Under the relaxed assumption the sink falls by a third to a half within 60 years, while the stock stays 21–40% below balance: the *sink* saturates long before the *stock*, because the fast pools approach balance first and the remaining distance lies in the slow pool. Saturation of the sink, not of the stock.

3.0.12.2 ⇝ **How much input would push saturation forward in time?** The models are linear in their input, so the projections with the litter held and with +20% answer it exactly for any target sink (`doublechecks/input_to_hold_sink.R`):

target sink	window	Yasso07	Yasso15	Yasso20	TP2 / TP3	SP1
observed 1985–2024 (+0.26)	10 yr	+8%	+7%	+5%	+5 / +5%	+11%
	20 yr	**+11%**	**+10%**	**+11%**	+7 / +6%	+13%
each model's own 1985–2024	20 yr	+9%	+7%	+3%	+10 / +10%	+4%
observed 2006–2024 (+0.12)	20 yr	+2%	+2%	+3%	−5 / −5%	+6%

About a tenth more input, sustained, holds the long-term sink for twenty years (0.4–0.5 $\mathrm{tC\,ha^{-1}\,yr^{-1}}$ of effective input in Yasso; each +10% buys 0.15–0.24 $\mathrm{tC\,ha^{-1}\,yr^{-1}}$ of sink).

3.0.12.3 ⇝ **A short digression: saturation comes regardless.** A higher input raises the eventual balance, so the soil gains again, but the new sink fades in the same way: with +20% from 2025 the versions of Yasso realise about half of the eventual gain within 60 years, and the gain lasts only while the input does. Input postpones saturation; it does not remove it. **A figure to build** from the forward runs already done: held litter against +20%, the sink fading in both.

Lower bounds: no further warming is in these projections. The required input depends on both the starting state (the baseline sink) and the transit time (the gain per unit of input).

Which target, the observed long-term rate, the recent one, or each model's own? A warming-corrected column for Yasso? National units ($\mathrm{Mt\,CO_2\,yr^{-1}}$) need a sourced area and Aleksi's eye.

3.0.13 Side arc b: the sensitivity to temperature

3.0.13.1 Relaxing the assumption changes the kinetics the calibration retains: the balanced-start calibration settles at faster and more temperature-sensitive decomposition (R1). Under warming to

SSP2-4.5 the stock falls by 5–7 tC ha^{-1} by 2084 in all three Yasso versions; only Yasso07 separates the published and calibrated parameters (-2.66 tC ha^{-1}), because its single climate modifier ties turnover to temperature sensitivity. For the inventory, which runs Yasso07, that is the applied point.
▶ F15_forward_arms

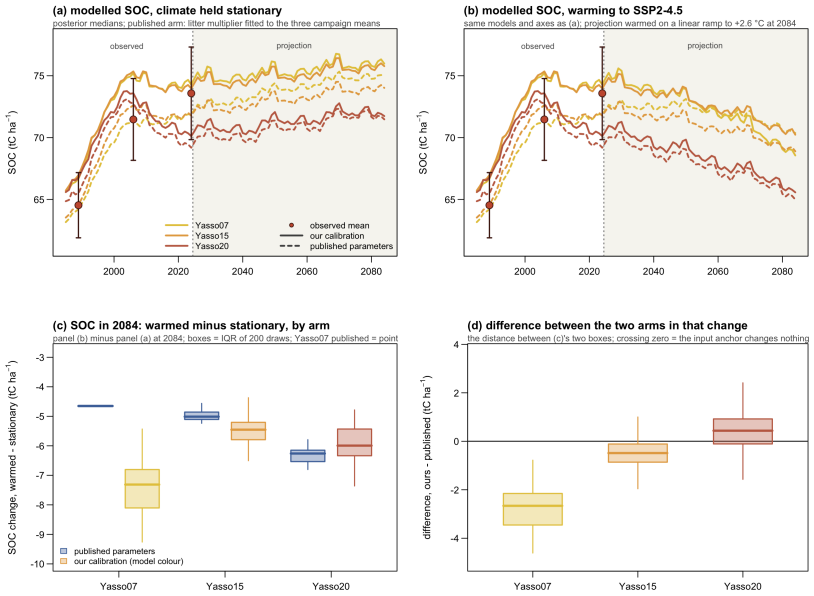
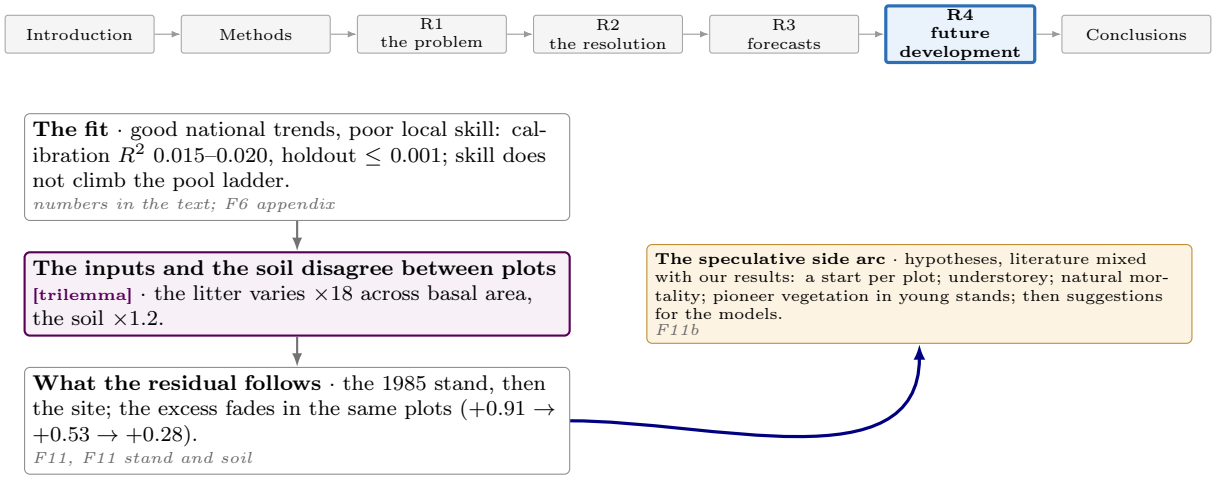


Figure 18: F15_forward_arms — main text

The comparison this arc needs is not yet computed. F15 compares the published and the calibrated parameters, both with a transient start; the arc as framed compares the two *assumptions*. A rerun of the forward experiment on the balanced-start posteriors would give the warming response with and without the history — cheap, no new calibration.

R4 — FUTURE DEVELOPMENT: WHY THE MODELS FAIL LOCALLY



Role (Lorenzo, 2026-10-13). A side arm, relevant but quite independent from the rest: an analysis of the implications of the model fit, ending with suggestions for the models.

3.0.14 Good trends, poor local skill

3.0.14.1 Calibration R^2 0.015–0.020, holdout $R^2 \leq 0.001$; skill does not climb the pool ladder. The models reproduce the national trend and almost none of the differences between plots. Numbers in the text; F6 in the appendix. ▶ F6_obs_vs_pred_holdout

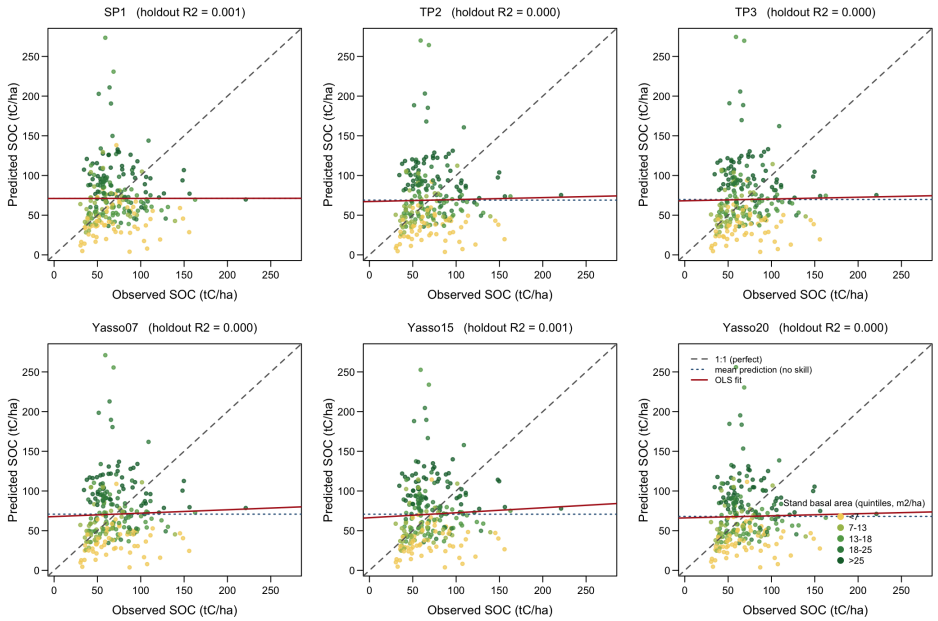


Figure 19: F6_obs_vs_pred_holdout — **appendix**

3.0.15 The inputs and the soil disagree between plots

3.0.15.1 The trilemma at plot scale. The litter product varies about ×18 across the basal-area range; the soil about ×1.2. A soil in balance with its litter would follow it — the inputs corner of the trilemma seen plot by plot, in one sentence.

3.0.16 What the residual follows: each plot carries its own history

3.0.16.1 The residual follows the 1985 stand, then the site (F11 → F11 stand and soil). In the same plots it fades over the three campaigns (+0.91 → +0.53 → +0.28 for sparse 1985 stands): the signature of a wrong starting state, not of a missing input. One national pre-run shape cannot carry each stand's history. ▶ F11_rf_heatmap

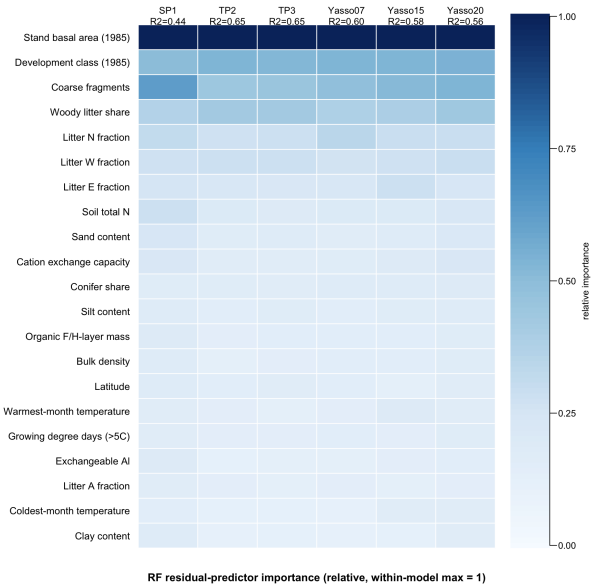


Figure 20: F11_rf_heatmap — main text

▶ F11_stand_and_soil

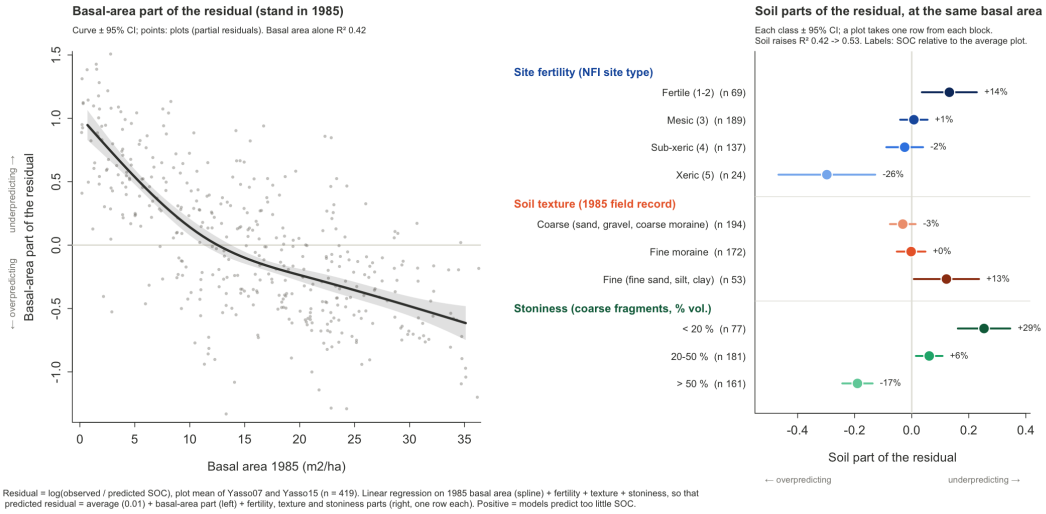


Figure 21: F11_stand_and_soil — main text

3.0.17 The speculative side arc: what the models miss

3.0.17.1 Hypotheses, with the literature mixed with our results.

1. **A starting state per plot.** The plot-level error fades at about the models' turnover pace: a start built from each plot's own history (the last regeneration cut dated from the 1985 stand age) is the natural next step. The excess lives in stands young in 1985. ▶ F11b_time_since_cut

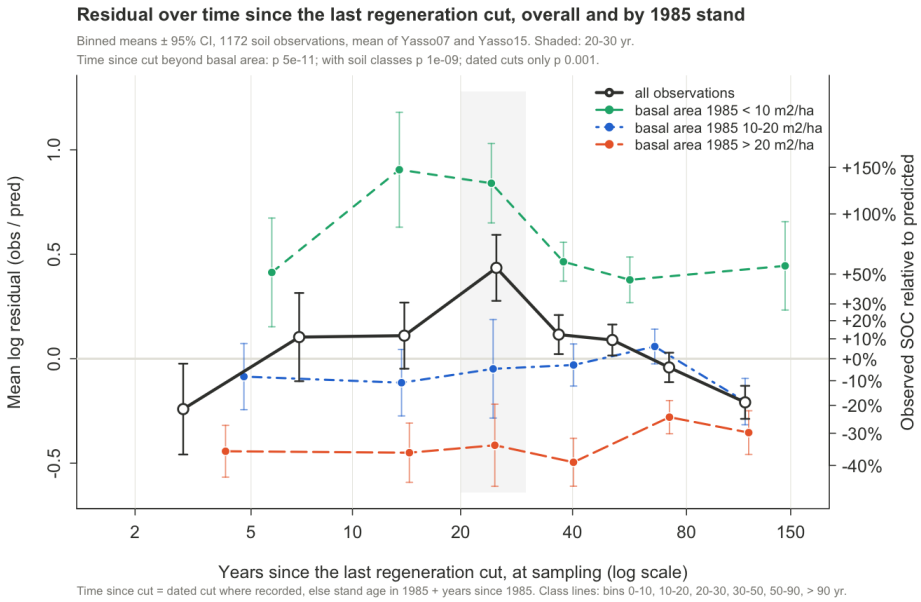


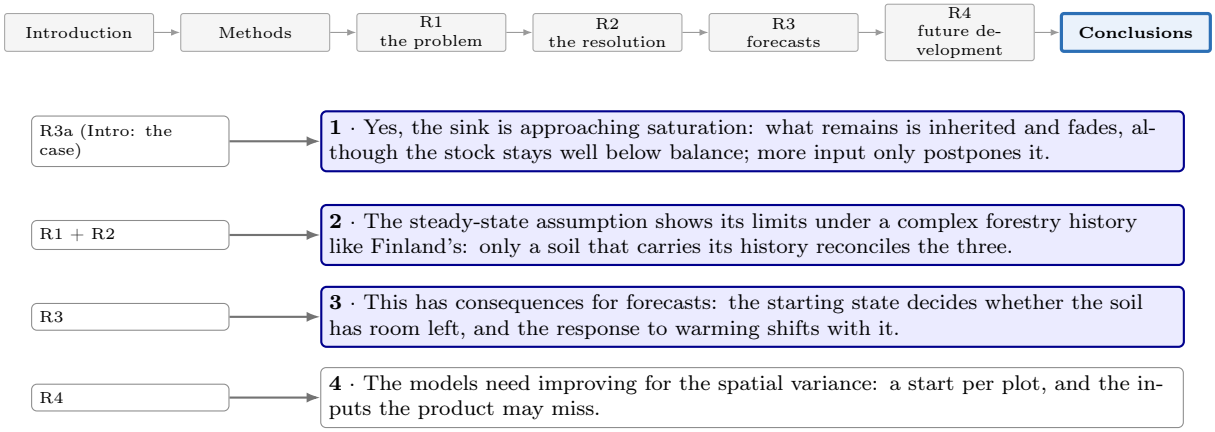
Figure 22: F11b_time_since_cut — main text

2. **Understorey.** Field-layer cover in 2023 (438 of 456 plots) adds a little after basal area; the plot-level understorey litter gives (J+u)/J ≈ 1.13, below the posteriors.
3. **Natural mortality** — quantified in the national inventory, 0.1–0.16 tC ha^{-1} yr^{-1} by now in the south, absent from the product; roots and rhizodeposition; mycorrhizal fungi; fine-root lifespan (pointing the other way). A set that brackets the input gap, never a sum that matches it.
4. **Pioneer vegetation, where the two meet.** The excess carbon of the stands young in 1985 was built during early stand development, either inherited or supplied by vegetation no tree-based product records; the data do not separate the two. Lauri Mehtätalo (LUKE) to be cited **[OPEN:** which work**]**.
5. **The site, briefly.** Fertility, texture and stones explain a further tenth of the residual **[OPEN:** check whether the stone effect is partly the stone correction of the observed stocks**]**.

3.0.18 Suggestions

3.0.18.1 A short list for the models, drawn from the above: a starting state per plot, built from each stand's dated history; an understorey and natural-mortality term in the litter input; an early-succession input after regeneration cuts; and the site (fertility, stones) where the litter alone cannot order the soil.

4 Conclusions



4.0.0.1 1. Yes, we are approaching saturation. Under the relaxed assumption the remaining sink is inherited and fades, by a third to a half within 60 years, although the stock stays 21–40% below balance: the sink saturates, not the stock. More input postpones saturation — about a tenth more holds the past sink for twenty years — but does not remove it.

4.0.0.2 2. The steady-state assumption shows its limits under a complex forestry history like Finland's. Started in balance, the models cannot reconcile the soil series, the kinetics and the inputs; carrying the history, they can, as far as the data allow, and the tension concentrates in the input corner.

4.0.0.3 3. This has consequences for forecasts. The starting state decides whether the soil has room left: a soil assumed balanced has none, one that carries its history keeps a fading sink. The response to warming shifts with the assumption where turnover and temperature sensitivity are tied (Yasso07).

4.0.0.4 4. The models need improving for the spatial variance. They reproduce the national trend but not the plots: a starting state per plot, and the inputs no tree-based product records.

Open questions collected

1. The weight budget for the recomposed story (15 / 30 / 30 / 25).
2. Merge F10b and F3_mean_SOCalt into one Introduction figure.
3. The 2021 source as an illustration in the Finnish case: keep, soften or move.
4. "Approaching saturation": of the sink, not the stock; and the observed slowing rests on the fragile interval — say "suggest" in the Introduction.
5. The balanced-start comparison named in the objectives, or only in the Methods.
6. The written expectations and the two that failed, in the Methods.
7. R3b: compute the warming response under the balanced-start calibration.
8. R3a: build the saturation figure (held litter vs +20%).
9. S16: five terms (decided) once the joint-effect runs are recalibrated; recalibrated three parts until then. The size of its climate part depends on the reference climate.

10. The target sink in the input table; a warming-corrected column; national units.
11. The recalibrated decomposition: DONE (2026-10-21), changed little. Decide whether the fixed-parameter version stays in the appendix.
12. Which appendix figures to promote: F19, the saturation figure, F4, F9.